

Dissertation Proposal One-Pager

AI/ML-Enabled Microstructure Analysis Pairing for Atom Probe Tomography and Time-of-Flight Secondary Ion Mass Spectrometry

Jose Tupayachi | September 18, 2026

Dissertation Theme

This dissertation develops and validates a shared multimodal machine-learning framework, comprising spectral and spatial encoders, cross-attention fusion, and LLM reporting, to automate ranging, reconstruction disambiguation, and microstructure feature extraction across two nanoscale characterization modalities: Atom Probe Tomography (APT) and Time-of-Flight Secondary Ion Mass Spectrometry (ToF-SIMS). The three papers form an escalating research approach: (1) Establish the shared framework and demonstrate the manual bottlenecks it addresses, (2) Automate two of the most challenging tasks, ranging and pole indexing, and (3) close the loop with literature supported reporting.

Background & Motivation

APT datasets encode two complementary, modalities: a mass spectrum requiring element/molecular-ion *ranging*, and a reconstructed 3D atom map from which precipitates, boundaries, clusters, and CSRO/CMRO are extracted.

Both steps are manual, expertise-dependent, and scale poorly on APT datasets. A demonstrated LAMDA workflow on real data confirms concrete bottlenecks: SEM-coregistered tip-profile calibration, reconstruction parameterization, pole-line indexing returning ambiguous BCC/FCC candidates, and RNG/RRNG ranging of overlapping molecular ions (e.g., Fe/Cr/Ni/Mo oxides and carbides). On the other hand, ToF-SIMS shares the same two-modality structure (spectrum + spatial ion image and accessible via pySPM). This motivates one shared AI/ML framework differing only in instrument-specific ingestion.

Figure 1. Overview of the proposed multimodal AI/ML framework linking spectral and spatial representations exemplifying APT analysis.

Three-Paper Structure

Paper 1 (Basis for Fall proposal defense): A Shared AI/ML Framework for Automated Ranging and Microstructure Screening in APT and ToF-SIMS

- Establishes the common two-modality representation and staged framework (spectral encoder + spatial encoder + fusion) shared by APT and ToF-SIMS.
- Documents the manual baseline and its concrete failure modes on real LAMDA APT data (reconstruction ambiguity, molecular-ion overlap).
- Defines the FAIR transcoding layer for vendor formats (POS/RNG/RRNG for APT, ITA/ITM/ITS via pySPM for ToF-SIMS) as shared ingestion.

Paper 2 (Year 2): Automated Multimodal Ranging and Reconstruction via ML-ToF/Bayesian Peak Assignment and Learned Pole Indexing

- 1D CNN/transformer spectral encoder for ranging, CNN/GNN spatial encoder for cluster and phase detection, resolves the BCC-vs-FCC pole-indexing ambiguity and molecular-ion overlap identified in Paper 1.

Paper 3 (Year 3): LLM-Driven Reporting and Literature-Grounded Interpretation for Correlative APT-ToF-SIMS Characterization

- Integrates fused spectral/spatial embeddings into a LLM + literature RAG reporting layer.
- Validates automated reports against expert-authored reports, benchmarks the deployed vLLM pipeline.

Committee Formation Plan

- Advisor/chair: Dr. Li & Dr. Yu & Dr. Yao & Dr. Das.

Timeline to Fall Proposal Defense

- **Now - 9/18:** This one-pager finalized.
- **Sept-Oct:** Finalize Paper 1 draft, internal review.
- **October:** Circulate written proposal to committee (≥ 2 weeks before defense).
- **November:** Schedule and defend proposal.

